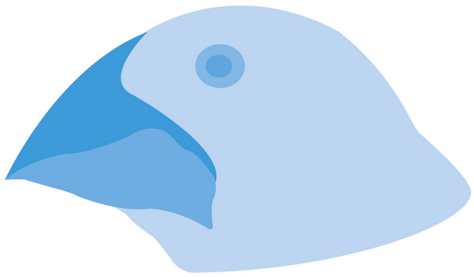
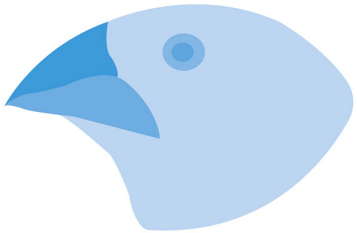
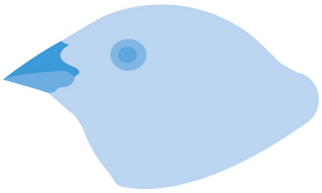
Comparison of Galápagos finch bill structure



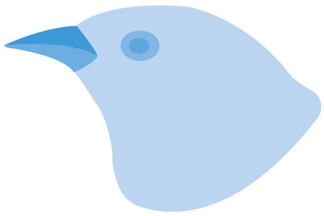
Geospiza magnirostris
The short, sharp bill of the Large Ground Finch, the biggest of Darwin's finches, enables it to crack nuts.



Geospiza fortis
The bill of the Medium Ground Finch is variable, evolving rapidly to adapt to whatever size seeds are available.



Geospiza parvula
The stubby bill of the Small Tree Finch, which forages in foliage, suits its diet of seeds, fruits, and insects.



Certhidea olivacea
The slender, probing bill of the Green Warbler-finch helps it catch small insects and spiders.

Read accessible description

Refining the theory

Darwin was influenced by Thomas Malthus's *An Essay on the Principle of Population* (1798), in which Malthus predicted that population growth would eventually outstrip food production. This idea matched the evidence Darwin had observed of ongoing competition between individual animals and species for resources. This competitive aspect formed the backbone of Darwin's coalescing theory of evolution.

By 1839, Darwin had developed an idea of evolution by natural selection. He was, though, reluctant to publish because he understood that the theory